

Exercise Set 2.6 (Page 97)

Evaluate the given limits (Problems 1 – 5):

1. $\lim_{(x,y) \rightarrow (0,0)} \frac{5-x^2}{4+x+y}$

Sol. $\lim_{(x,y) \rightarrow (0,0)} \frac{5-x^2}{4+x+y} = \frac{\lim_{(x,y) \rightarrow (0,0)} (5-x^2)}{\lim_{(x,y) \rightarrow (0,0)} (4+x+y)} = \frac{5-0}{4+0+0} = \frac{5}{4}$

2. $\lim_{(x,y) \rightarrow (1,-1)} e^{-xy}$

Sol. $\lim_{(x,y) \rightarrow (1,-1)} e^{-xy} = e^{-(1)(-1)} = e$

3. $\lim_{(x,y) \rightarrow (0,0)} \frac{e^{xy} \sin xy}{xy}$

Sol. $\lim_{(x,y) \rightarrow (0,0)} \frac{e^{xy} \sin xy}{xy} = \lim_{(x,y) \rightarrow (0,0)} e^{xy} \cdot \lim_{(x,y) \rightarrow (0,0)} \frac{\sin xy}{xy} = 1$

4. $\lim_{(x,y) \rightarrow (0,0)} \frac{x^3 - y^3}{x^2 + y^2}$

Sol. Setting $x = r \cos \theta$, $y = r \sin \theta$, we have the given limit

$$= \lim_{r \rightarrow 0} \frac{r^3 (\cos^3 \theta - \sin^3 \theta)}{r^2}, \text{ since } r^2 = x^2 + y^2$$

and $(x,y) \rightarrow (0,0)$ implies $r \rightarrow 0$

$$= \lim_{r \rightarrow 0} r (\cos^3 \theta - \sin^3 \theta) = 0$$

5. $\lim_{(x,y) \rightarrow (2,2)} \frac{x^3 - 2xy + 3x^2 - 2y}{x^2y + 4y^2 - 6x^2 + 24y}$

Sol. $\lim_{(x,y) \rightarrow (2,2)} \frac{x^3 - 2xy + 3x^2 - 2y}{x^2y + 4y^2 - 6x^2 + 24y}$

$$= \frac{\lim_{(x,y) \rightarrow (2,2)} (x^3 - 2xy + 3x^2 - 2y)}{\lim_{(x,y) \rightarrow (2,2)} (x^2y + 4y^2 - 6x^2 + 24y)}$$

$$= \frac{8 - 8 + 12 - 4}{8 + 16 - 24 + 48} = \frac{8}{48} = \frac{1}{6}$$

In each of Problems 6 – 10, show that the given limit does not exist:

6. $\lim_{(x,y) \rightarrow (0,0)} \frac{xy}{x^2 + y^2}$

Sol. The limit does not exist if we show that $f(x, y) = \frac{xy}{x^2 + y^2}$ approaches to different values as $(x, y) \rightarrow (0, 0)$ along different paths. Let (x, y) approach $(0, 0)$ along the line $y = mx$.

Then $f(x, y) = \frac{mx^2}{x^2(1 + m^2)} = \frac{m}{1 + m^2}$, since $x \neq 0$

Thus $f(x, y)$ will have different values for different values of m . Hence the given limit does not exist.

7. $\lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2}$

Sol. Let $x = r \cos \theta$, $y = r \sin \theta$. Then $r^2 = x^2 + y^2$.

As $(x, y) \rightarrow (0, 0)$, $r \rightarrow 0$. Therefore,

$$\begin{aligned} \lim_{(x,y) \rightarrow (0,0)} \frac{x^2 - y^2}{x^2 + y^2} &= \lim_{(x,y) \rightarrow (0,0)} \frac{r^2(\cos^2 \theta - \sin^2 \theta)}{r^2} \\ &= \lim_{r \rightarrow 0} (\cos^2 \theta - \sin^2 \theta) = \cos 2\theta \end{aligned}$$

which is independent of r and can have any value. Thus the given limit does not exist.

8. $\lim_{(x,y) \rightarrow (0,0)} \frac{ax^2 + by}{cy^2 + dx}$

Sol. Let $(x, y) \rightarrow (0, 0)$ along the line $y = mx$

$$\begin{aligned} \lim_{(x,y) \rightarrow (0,0)} \frac{ax^2 + by}{cy^2 + dx} &= \lim_{(x,y) \rightarrow (0,0)} \frac{ax^2 + bmx}{cm^2x^2 + dx} \\ &= \lim_{(x,y) \rightarrow (0,0)} \frac{ax + bm}{cm^2x + d} \\ &= \frac{bm}{d}, \text{ which has different values for} \end{aligned}$$

different m . Hence the given limit does not exist.

9. $\lim_{(x,y) \rightarrow (0,0)} \frac{(x^2 + y^2)^2}{x^4 + y^4}$

Sol. Let $(x, y) \rightarrow (0, 0)$ along the line $ax + by = 0$. Then

$$\begin{aligned} \lim_{(x,y) \rightarrow (0,0)} \frac{(x^2 + y^2)^2}{x^4 + y^4} &= \lim_{(x,y) \rightarrow (0,0)} \frac{\left(x^2 + \frac{a^2}{b^2}x^2\right)}{x^4 + \frac{a^4}{b^4}x^4} \\ &= \lim_{(x,y) \rightarrow (0,0)} \frac{(a^2 + b^2)^2}{a^4 + b^4} = \frac{(a^2 + b^2)^2}{a^4 + b^4} \end{aligned}$$

which depends on the values of a and b .
Hence the limit does not exist.

10. $\lim_{(x,y) \rightarrow (0,0)} \left(\frac{xy^2}{x^2 + y^4} \right)$

Sol. Let $(x, y) \rightarrow (0, 0)$ along $y = mx$. Then

$$f(x, y) = \frac{xy^2}{x^2 + y^4} = \frac{m^2x^3}{x^2 + m^4x^4} = \frac{m^2x}{1 + m^4x^2}$$

$$\frac{m^2x}{1 + m^4x^2} \rightarrow 0 \text{ as } x \rightarrow 0$$

Thus along every straight line through the origin,

$$f(x, y) \rightarrow 0 \text{ as } (x, y) \rightarrow (0, 0)$$

Next, let $(x, y) \rightarrow (0, 0)$ along $x = y^2$. Then

$$f(x, y) = \frac{y^4}{y^4 + y^4} = \frac{1}{2}, \text{ so that } f(x, y) \rightarrow \frac{1}{2} \text{ as } (x, y) \rightarrow (0, 0)$$

along the parabola $x = y^2$. Hence the limit does not exist.

11. Let $f(x, y) = \begin{cases} \frac{xy^2}{x^3 + y^3} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$

Show that f is not continuous at the origin.

Sol. We evaluate $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$.

Let $(0, 0)$ be approached along the line $y = mx$

$$\lim_{(x,y) \rightarrow (0,0)} \frac{x^2y^2}{x^3 + y^3} = \lim_{(x,y) \rightarrow (0,0)} \frac{m^2x^3}{x^3 + m^3x^3}$$

$$= \frac{m^2}{1 + m^3}, \text{ since } x \neq 0,$$

which will have different values for different m . Therefore,
 $\lim_{(x,y) \rightarrow (0,0)} f(x, y)$ does not exist. Hence the function is not continuous at $(0, 0)$.

12. Find a such that the function

$$f(x, y) = \begin{cases} \frac{3xy}{\sqrt{x^2 + y^2}} & \text{if } (x, y) \neq (0, 0) \\ a & \text{if } (x, y) = (0, 0) \end{cases}$$

is continuous at $(0, 0)$.

Sol. Let $x = r \cos \theta$, $y = r \sin \theta$. Then $r \rightarrow 0$ as $(x, y) \rightarrow (0, 0)$

$$f(x, y) = \frac{3r^2 \sin \theta \cos \theta}{r} = 3r \sin \theta \cos \theta$$

$$3r \sin \theta \cos \theta \rightarrow 0 \text{ as } r \rightarrow 0.$$

Thus $\lim_{(x,y) \rightarrow (0,0)} f(x, y) = 0$

The function will be continuous at the origin if $f(0, 0) = 0$ which requires $a = 0$.

$$13. \text{ Let } f(x, y) = \begin{cases} \frac{x^3 + y^3}{x^2 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Examine the continuity at $(0, 0)$. Do $f_x(0, 0)$ and $f_y(0, 0)$ exist?

Sol. Let $x = r \cos \theta$, $y = r \sin \theta$. Then $r \rightarrow 0$ as $(x, y) \rightarrow (0, 0)$

$$f(x, y) = \frac{r^3(\cos^3 \theta + \sin^3 \theta)}{r^2} = r(\cos^3 \theta + \sin^3 \theta)$$

$$r(\cos^3 \theta + \sin^3 \theta) \rightarrow 0 \text{ as } r \rightarrow 0.$$

Therefore, f is continuous at $(0, 0)$.

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h - 0}{h} = 1$$

$$f_y(0, 0) = \lim_{k \rightarrow 0} \frac{f(0, k) - f(0, 0)}{k}$$

$$= \lim_{k \rightarrow 0} \frac{k - 0}{k} = 1$$

Thus $f_x(0, 0)$ and $f_y(0, 0)$ exist.

$$14. \text{ Let } f(x, y) = \begin{cases} \frac{x^2 y}{x^4 + y^2} & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$$

Prove that f is not continuous at $(0, 0)$. Do $f_x(0, 0)$ and $f_y(0, 0)$ exist?

Sol. Let $(0, 0)$ be approached along the line $y = mx$

$$f(x, y) = \frac{mx^3}{x^4 + m^2 x^2} = \frac{mx}{x^2 + m^2} \rightarrow 0 \text{ as } x \rightarrow 0.$$

But if we approach $(0, 0)$ along $y = x^2$, then

$$f(x, y) = \frac{x^4}{x^4 + x^4} = \frac{1}{2} \text{ and so } \lim_{(x, y) \rightarrow (0, 0)} f(x, y) = \frac{1}{2}$$

Thus $\lim_{(x, y) \rightarrow (0, 0)} f(x, y)$ does not exist.

Hence f is not continuous at $(0, 0)$.

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0 - 0}{h} = 0$$

$$f_y(0, 0) = \frac{f(0, k) - f(0, 0)}{k} = \lim_{k \rightarrow 0} \frac{0 - 0}{k} = 0$$

Thus $f_x(0, 0) = f_y(0, 0) = 0$.

Find the first order derivatives of the given functions (Problems 15 – 22):

15. $f(x, y) = x^{y^2}$

Sol. $f(x, y) = x^{y^2}$

$$f_x = y^2 \cdot x^{y^2-1} ; f_y = 2y \cdot x^{y^2} \ln x, \left[\frac{d}{dx} (a^x) = a^x \ln a \right]$$

16. $f(x, y) = e^{x^2+y^2}$

Sol. $f(x, y) = e^{x^2+y^2}$

$$f_x = (e^{x^2+y^2}) 2x = 2xe^{x^2+y^2} ; f_y = (e^{x^2+y^2}) 2y = 2ye^{x^2+y^2}$$

17. $f(x, y) = \arctan\left(\frac{y}{x}\right)$

Sol. $f(x, y) = \arctan\left(\frac{y}{x}\right)$

$$f_x = \frac{1}{1 + \left(\frac{y}{x}\right)^2} \cdot \frac{\partial}{\partial x} \left(\frac{y}{x}\right) = \frac{1}{1 + \left(1 + \frac{y}{x}\right)} \cdot \left(-\frac{y}{x^2}\right)$$

$$= -\frac{1}{x^2 \left\{ 1 + \left(\frac{y}{x}\right)^2 \right\}} = -\frac{y}{x^2 \left\{ \left(\frac{x^2 + y^2}{x^2}\right) \right\}} = -\frac{y}{x^2 + y^2}$$

$$f_y = \frac{1}{1 + \left(\frac{y}{x}\right)^2} \cdot \frac{\partial}{\partial y} \left(\frac{y}{x}\right) = \frac{1}{x^2 + y^2} \cdot \frac{1}{x} = \frac{x^2}{x^2 + y^2} \cdot \frac{1}{x} = \frac{x}{x^2 + y^2}$$

18. $f(x, y) = \arctan(x + y)$

Sol. Let $z = \arctan(x + y)$

$$\text{Then } \frac{\partial z}{\partial x} = \frac{1}{1 + (x + y)^2} \cdot 1 = \frac{1}{1 + (x + y)^2}$$

$$\text{and } \frac{\partial z}{\partial y} = \frac{1}{1 + (x + y)^2} \cdot 1 = \frac{1}{1 + (x + y)^2}$$

19. $f(x, y) = e^{ax} \sin by$

Sol. Let $z = e^{ax} \sin by$

$$\text{Then } \frac{\partial z}{\partial x} = (e^{ax} \cdot a) \sin by = a e^{ax} \sin by$$

$$\text{and } \frac{\partial z}{\partial y} = e^{ax} (\cos by) b = b e^{ax} \cos by$$

20. $f(x, y) = \ln(x^2 + y^2)$

Sol. Let $z = \ln(x^2 + y^2)$

$$\frac{\partial z}{\partial x} = \frac{1}{x^2 + y^2} \cdot 2x = \frac{2x}{x^2 + y^2}$$

$$\frac{\partial z}{\partial y} = \frac{1}{x^2 + y^2} \cdot 2y = \frac{2y}{x^2 + y^2}$$

$$21. f(x, y) = \ln \left[\frac{\sqrt{x^2 + y^2} - x}{\sqrt{x^2 + y^2} + x} \right]$$

$$\begin{aligned} \text{Sol. } f(x, y) &= \ln \left\{ \frac{\sqrt{x^2 + y^2} - x}{\sqrt{x^2 + y^2} + x} \right\} \\ &= \ln \{ \sqrt{x^2 + y^2} - x \} - \ln \{ \sqrt{x^2 + y^2} + x \} \\ f_x &= \frac{1}{\sqrt{x^2 + y^2} - x} \cdot \frac{\partial}{\partial x} (\sqrt{x^2 + y^2} - x) \\ &\quad - \frac{1}{\sqrt{x^2 + y^2} + x} \cdot \frac{\partial}{\partial x} (\sqrt{x^2 + y^2} + x) \\ &= \frac{1}{\sqrt{x^2 + y^2} - x} \cdot \left\{ \frac{1}{2} (x^2 + y^2)^{-1/2} (2x) - 1 \right\} \\ &\quad - \frac{1}{\sqrt{x^2 + y^2} + x} \cdot \left\{ \frac{1}{2} (x^2 + y^2)^{-1/2} (2x) + 1 \right\} \\ &= \frac{1}{\sqrt{x^2 + y^2} - x} \cdot \left\{ \frac{x}{(x^2 + y^2)^{1/2}} - 1 \right\} \\ &\quad - \frac{1}{\sqrt{x^2 + y^2} + x} \cdot \left\{ \frac{x}{(x^2 + y^2)^{1/2}} + 1 \right\} \\ &= \frac{1}{\sqrt{x^2 + y^2} - x} \cdot \left\{ \frac{x - (x^2 + y^2)^{1/2}}{(x^2 + y^2)^{1/2}} \right\} \\ &\quad - \frac{1}{\sqrt{x^2 + y^2} + x} \cdot \left\{ \frac{x + (x^2 + y^2)^{1/2}}{(x^2 + y^2)^{1/2}} \right\} \\ &= \frac{-1}{(x^2 + y^2)^{1/2}} - \frac{1}{(x^2 + y^2)^{1/2}} = \frac{-2}{\sqrt{x^2 + y^2}} \\ f_y &= \frac{1}{\sqrt{x^2 + y^2} - x} \cdot \frac{\partial}{\partial y} (\sqrt{x^2 + y^2} - x) \\ &\quad - \frac{1}{\sqrt{x^2 + y^2} + x} \cdot \frac{\partial}{\partial y} (\sqrt{x^2 + y^2} + x) \\ &= \frac{1}{\sqrt{x^2 + y^2} - x} \cdot \left\{ \frac{1}{2} (x^2 + y^2)^{-1/2} \cdot (2y) \right\} \\ &\quad - \frac{1}{\sqrt{x^2 + y^2} + x} \cdot \left\{ \frac{1}{2} (x^2 + y^2)^{-1/2} \cdot (2y) \right\} \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{\sqrt{x^2 + y^2} - x} \cdot \left(\frac{y}{\sqrt{x^2 + y^2}} \right) - \frac{1}{\sqrt{x^2 + y^2} + x} \cdot \left(\frac{y}{\sqrt{x^2 + y^2}} \right) \\
&= \frac{y}{\sqrt{x^2 + y^2}} \cdot \left\{ \frac{\sqrt{x^2 + y^2} + x - \sqrt{x^2 + y^2} + x}{(\sqrt{x^2 + y^2} - x)(\sqrt{x^2 + y^2} + x)} \right\} \\
&= \frac{y}{\sqrt{x^2 + y^2}} \cdot \left(\frac{2x}{x^2 + y^2 - x^2} \right) \\
&= \frac{2xy}{y^2 \sqrt{x^2 + y^2}} = \frac{2x}{y \sqrt{x^2 + y^2}}
\end{aligned}$$

22. $f(x, y, z) = \frac{1}{\sqrt{x^2 + y^2 + z^2}}$

Sol. $f(x, y, z) = \frac{1}{\sqrt{x^2 + y^2 + z^2}} = (x^2 + y^2 + z^2)^{-1/2}$

Now, $f_x = -\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} \frac{\partial}{\partial x} (x^2 + y^2 + z^2)$
 $= -\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} \cdot (2x) = \frac{-x}{(x^2 + y^2 + z^2)^{3/2}}$

$f_y = -\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} \frac{\partial}{\partial y} (x^2 + y^2 + z^2)$
 $= -\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} (2y) = \frac{-y}{(x^2 + y^2 + z^2)^{3/2}}$

$f_z = -\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} \frac{\partial}{\partial z} (x^2 + y^2 + z^2)$
 $= -\frac{1}{2} (x^2 + y^2 + z^2)^{-3/2} (2z) = \frac{-z}{(x^2 + y^2 + z^2)^{3/2}}$

Find all the four second order partial derivatives (Problems 23 – 26):

23. e^{x-y}

Sol. Let $z = e^{x-y}$ Then

$$\frac{\partial z}{\partial x} = e^{x-y} \cdot 1 = e^{x-y}$$

$$\frac{\partial^2 z}{\partial x^2} = e^{x-y} \cdot 1 = e^{x-y}$$

and $\frac{\partial z}{\partial y} = e^{x-y} \cdot (-1) = -e^{x-y}$

$$\frac{\partial^2 z}{\partial y^2} = (-e^{x-y}) \cdot (-1) = e^{x-y}$$

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right) = \frac{\partial}{\partial x} (-e^{x-y}) = -e^{x-y} \cdot 1 = -e^{x-y}$$

$$\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial}{\partial y} \right) \left(\frac{\partial z}{\partial x} \right) (e^{x-y}) = e^{x-y} \cdot -1 = -e^{x-y}$$

24. $\frac{x+y}{x-y}$

Sol. Let $z = \frac{x+y}{x-y}$

$$\frac{\partial z}{\partial x} = \frac{(x-y) \cdot 1 - (x+y) \cdot 1}{(x-y)^2} = \frac{-2y}{(x-y)^2}$$

$$\frac{\partial^2 z}{\partial x^2} = (-2y) \cdot (-2) \cdot \frac{1}{(x-y)^3} \cdot 1 = \frac{4y}{(x-y)^3}$$

$$\begin{aligned} \frac{\partial^2 z}{\partial y \partial x} &= \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right) = \frac{\partial}{\partial y} \left(\frac{-2y}{(x-y)^2} \right) \\ &= \frac{(x-y)^2 (-2) - (-2y) (2) (x-y) (-1)}{(x-y)^4} \\ &= \frac{-2(x-y)[x-y+2y]}{(x-y)^4} = \frac{-2(x+y)}{(x-y)^3} \end{aligned}$$

Also $\frac{\partial z}{\partial y} = \frac{(x-y)(1) - (x+y)(-1)}{(x-y)^2} = \frac{2x}{(x-y)^2}$

$$\frac{\partial^2 z}{\partial y^2} = (2x)(x-y)^{-3}(-2)(-1) = \frac{4x}{(x-y)^3}$$

$$\begin{aligned} \frac{\partial^2 z}{\partial x \partial y} &= \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right) = \frac{\partial}{\partial x} \left(\frac{2x}{(x-y)^2} \right) \\ &= \frac{(x-y)^2 \cdot 2 - 2x \cdot 2(x-y) \cdot 1}{(x-y)^4} \\ &= \frac{2(x-y)[x-y-2x]}{(x-y)^4} \\ &= \frac{2(-x-y)}{(x-y)^3} = \frac{-2(x+y)}{(x-y)^3} \end{aligned}$$

25. e^{xy}

Sol. Let $u = e^{xy}$. Then

$$\frac{\partial u}{\partial x} = e^{xy} \times yx^{y-1} = yx^{y-1} e^{xy}$$

$$\frac{\partial u}{\partial y} = e^{xy} \frac{\partial}{\partial y} (x^y) = e^{xy} x^y \ln x$$

$$\frac{\partial^2 u}{\partial x^2} = y(y-1)x^{y-2} e^{xy} + yx^{y-1} \cdot e^{xy} \cdot (yx^{y-1})$$

$$= [y(y-1)x^{y-2} + y^2 x^{2y-2}] e^{xy}$$

$$\begin{aligned}
\frac{\partial^2 u}{\partial y \partial x} &= \frac{\partial}{\partial y} (y x^{y-1} e^{xy}) \\
&= x^{y-1} e^{xy} \frac{\partial}{\partial x} (y) + y x^{y-1} \frac{\partial}{\partial x} (e^{xy}) + y e^{xy} \frac{\partial}{\partial x} (x^{y-1}) \\
&= x^{y-1} e^{xy} + y x^{y-1} e^{xy} (x^y \ln x) + y e^{xy} x^{y-1} \ln x \\
&= e^{xy} (x^{y-1} + y x^{2y-1} \ln x + y x^{y-1} \ln x) \\
\frac{\partial^2 u}{\partial y^2} &= \frac{\partial}{\partial y} (e^{xy} x^y \ln x) = e^{xy} \frac{\partial}{\partial x} (x^y) \ln x + \frac{\partial}{\partial x} (e^{xy}) x^y \ln x \\
&= e^{xy} x^y (\ln x)^2 + e^{xy} x^y \ln x (x^y \ln x) \\
&= e^{xy} [x^y (\ln x)^2 + x^{2y} (\ln x)^2] \\
\frac{\partial^2 u}{\partial x \partial y} &= \frac{\partial}{\partial x} (e^{xy} x^y \ln x) \\
&= e^{xy} x^y \frac{\partial}{\partial x} (\ln x) + e^{xy} \ln x \frac{\partial}{\partial x} (x^y) + x^y \ln x \frac{\partial}{\partial x} (e^{xy}) \\
&= e^{xy} x^y \times \frac{1}{x} + e^{xy} \ln x y x^{y-1} + x^y \ln x e^{xy} y x^{y-1} \\
&= e^{xy} x^{y-1} + e^{xy} y \ln x x^{y-1} + e^{xy} (x^{2y-1} y \ln x) \\
&= e^{xy} [x^{y-1} + y x^{y-1} \ln x + x^{2y-1} y \ln x]
\end{aligned}$$

26. $\tan(\arctan x + \arctan y)$

Sol. Let $z = \tan(\arctan x + \arctan y)$

$$\begin{aligned}
z &= \frac{\tan \arctan x + \tan \arctan y}{1 - \tan(\arctan x) \tan(\arctan y)} = \frac{x + y}{1 - xy} \\
\frac{\partial z}{\partial x} &= \frac{(1 - xy) \cdot 1 - (x + y)(-y)}{(1 - xy)^2} = \frac{1 + y^2}{(1 - xy)^2} \\
\frac{\partial z}{\partial y} &= \frac{(1 - xy) \cdot 1 - (x + y)(-x)}{(1 - xy)^2} = \frac{1 + x^2}{(1 - xy)^2} \\
\frac{\partial^2 z}{\partial x^2} &= (1 + y^2) [-2(1 - xy)^{-3} (-y)] = \frac{2y(1 + y^2)}{(1 - xy)^3} \\
\frac{\partial^2 z}{\partial y^2} &= (1 + x^2) [-2(1 - xy)^{-3} \times (-x)] = \frac{2x(1 + x^2)}{(1 - xy)^3} \\
\frac{\partial^2 z}{\partial y \partial x} &= \frac{\partial}{\partial y} \left[\frac{1 + y^2}{(1 - xy)^2} \right] \\
&= \frac{(1 - xy)^2 \cdot 2y - (1 + y^2) \cdot 2(1 - xy)(-x)}{(1 - xy)^4} \\
&= \frac{2y(1 - xy)^2 + 2x(1 + y^2)(1 - xy)}{(1 - xy)^4} \\
&= \frac{2y(1 - xy) + 2x(1 + y^2)}{(1 - xy)^3} = \frac{2(x + y)}{(1 - xy)^3}
\end{aligned}$$

Similarly, or by symmetry (students are advised to work out themselves)

$$\frac{\partial^2 z}{\partial x \partial y} = \frac{2(y+x)}{(1-xy)^3}$$

In Problems 27 – 32 verify that $f_{xy} = f_{yx}$:

27. $f(x, y) = e^{xy} \cos(bx + c)$

Sol. $f_x = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(e^{xy}) \cos(bx + c) + e^{xy} \frac{\partial}{\partial x} \cos(bx + c)$
 $= ye^{xy} \cos(bx + c) - be^{xy} \sin(bx + c)$
 $f_{xy} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \cos(bx + c) \left\{ y \frac{\partial}{\partial y}(e^{xy}) + e^{xy} \frac{\partial}{\partial y}(y) \right\}$
 $\quad - b \sin(bx + c) \frac{\partial}{\partial y}(e^{xy})$
 $= \cos(bx + c) \{ xy e^{xy} + e^{xy} \} - b \sin(bx + c) x e^{xy}$
 or $f_{xy} = \frac{\partial^2 f}{\partial y \partial x} = xy e^{xy} \cos(bx + c) + e^{xy} \cos(bx + c)$
 $\quad - bx e^{xy} \sin(bx + c) \quad (1)$
 $f_y = \frac{\partial f}{\partial y} = x e^{xy} \cos(bx + c)$
 $f_{yx} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = x e^{xy} \frac{\partial}{\partial x} \cos(bx + c) + x \cos(bx + c) \frac{\partial}{\partial x}(e^{xy})$
 $\quad + e^{xy} \cos(bx + c) \frac{\partial}{\partial x}(x)$
 $= -bx e^{xy} \sin(bx + c) + xy \cos(bx + c) e^{xy} + e^{xy} \cos(bx + c)$
 $= xy e^{xy} \cos(bx + c) + e^{xy} \cos(bx + c) - bx e^{xy} \sin(bx + c) \quad (2)$

From (1) and (2), we see that $f_{xy} = f_{yx}$

28. $f(x, y) = \ln(e^x + e^y)$

Sol. $f(x, y) = \ln(e^x + e^y)$

$$f_y = \frac{\partial f}{\partial y} = \frac{1}{e^x + e^y} \frac{\partial}{\partial y}(e^x + e^y)$$

$$= \frac{1}{e^x + e^y} \cdot e^y = \frac{e^y}{e^x + e^y}$$

$$f_{yx} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{(e^x + e^y) \cdot 0 - e^y (e^x)}{(e^x + e^y)^2} = -\frac{e^x e^y}{(e^x + e^y)^2}$$

or $f_{yx} = \frac{\partial^2 f}{\partial x \partial y} = -\frac{e^{x+y}}{(e^x + e^y)^2} \quad (1)$

Now $f_x = \frac{\partial f}{\partial x} = \frac{1}{(e^x + e^y)} \cdot \frac{\partial}{\partial x}(e^x + e^y) = \frac{1}{e^x + e^y} = \frac{e^x}{e^x + e^y}$

$$f_{xy} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{(e^x + e^y) \cdot 0 - e^x \cdot e^y}{(e^x + e^y)^2} = -\frac{e^x e^y}{(e^x + e^y)^2}$$

$$\text{or } f_{xy} = \frac{\partial^2 f}{\partial x \partial y} = -\frac{e^{x+y}}{(e^x + e^y)^2} \quad (2)$$

Hence (1) and (2), we get $f_{xy} = f_{yx}$

$$29. f(x, y) = \ln \left(\frac{x^2 + y^2}{xy} \right)$$

$$\text{Sol. } f(x, y) = \ln \left(\frac{x^2 + y^2}{xy} \right) = \ln(x^2 + y^2) - \ln(xy)$$

$$\begin{aligned} \frac{\partial f}{\partial y} &= \frac{1}{x^2 + y^2} \cdot \frac{\partial}{\partial x}(x^2 + y^2) - \frac{1}{xy} \cdot \frac{\partial}{\partial x}(xy) \\ &= \frac{1}{x^2 + y^2} (2y) - \frac{1}{xy} \cdot (x) = \frac{2y}{x^2 + y^2} - \frac{x}{xy} = \frac{2y}{x^2 + y^2} - \frac{1}{y} \end{aligned}$$

$$\begin{aligned} f_{yx} &= \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{(x^2 + y^2) \frac{\partial}{\partial x}(2y) - (2y) \frac{\partial}{\partial x}(x^2 + y^2)}{(x^2 + y^2)^2} - \frac{\partial}{\partial x} \left(\frac{1}{y} \right) \\ &= \frac{(x^2 + y^2)(0) - 2y(2x)}{(x^2 + y^2)^2} - 0 = -\frac{4xy}{(x^2 + y^2)^2} \end{aligned}$$

$$\text{or } f_{yx} = \frac{\partial^2 f}{\partial x \partial y} = -\frac{4xy}{(x^2 + y^2)^2} \quad (1)$$

$$\begin{aligned} f_x &= \frac{1}{(x^2 + y^2)} \frac{\partial}{\partial x}(x^2 + y^2) - \frac{1}{(xy)} \frac{\partial}{\partial x}(xy) \\ &= \frac{1}{x^2 + y^2} (2x) - \frac{1}{(xy)} \cdot y = \frac{2x}{x^2 + y^2} - \frac{1}{x} \end{aligned}$$

$$\begin{aligned} f_{xy} &= \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{(x^2 + y^2) \frac{\partial}{\partial y}(2x) - (2x) \frac{\partial}{\partial y}(x^2 + y^2)}{(x^2 + y^2)^2} - \frac{\partial}{\partial y} \left(\frac{1}{x} \right) \\ &= \frac{(x^2 + y^2)0 - (2x)(0 + 2y)}{(x^2 + y^2)^2} - 0 = -\frac{4xy}{(x^2 + y^2)^2} \end{aligned}$$

$$\text{or } f_{xy} = \frac{\partial^2 f}{\partial y \partial x} = -\frac{4xy}{(x^2 + y^2)^2} \quad (2)$$

From (1) and (2), we get $f_{xy} = f_{yx}$

$$30. f(x, y) = x^y + y^x$$

$$\text{Sol. } f(x, y) = x^y + y^x$$

$$f_y = \frac{\partial f}{\partial y} = \frac{\partial}{\partial y}(x^y) + \frac{\partial}{\partial y}(y^x) = x^y \ln x + xy^{x-1}$$

$$f_{yx} = \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = x^y \frac{\partial}{\partial x}(\ln x) + \ln x \frac{\partial}{\partial x}(x^y) + x \frac{\partial}{\partial x}(y^{x-1}) + y^{x-1} \frac{\partial}{\partial x}(x)$$

$$= x^{y-1} + yx^{y-1} \ln x + xy^{x-1} \ln y + y^{x-1}$$

$$\text{or } f_{yx} = \frac{\partial^2 f}{\partial x \partial y} = x^{y-1} (1 + y \ln x) + y^{x-1} (1 + x \ln y) \quad (1)$$

$$\text{Now } f_x = \frac{\partial f}{\partial x} = \frac{\partial}{\partial x} (x^y) + \frac{\partial}{\partial x} (y^x) = yx^{y-1} + y^x \ln y \cdot 1$$

$$\begin{aligned} f_{xy} &= \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = y \cdot \frac{\partial}{\partial y} (x^{y-1}) + (x^{y-1}) \frac{\partial}{\partial y} (y) + y^x \frac{\partial}{\partial y} (\ln y) + \ln y \frac{\partial}{\partial y} (y^x) \\ &= y \cdot x^{y-1} \cdot \ln x + x^{y-1} + y^x \cdot \frac{1}{y} + \ln y (x \cdot y^{x-1}) \\ &= yx^{y-1} \ln x + x^{y-1} + y^{x-1} + xy^{x-1} \ln y \\ &= x^{y-1} (y \ln x + 1) + y^{x-1} (1 + x \ln y) \end{aligned} \quad (2)$$

From (1) and (2), we get $f_{xy} = f_{yx}$

$$31. f(x, y) = x \sin xy + y \cos xy$$

$$\text{Sol. } f_x = \sin xy + xy \cos xy - y^2 \sin xy$$

$$\begin{aligned} f_{xy} &= x \cos xy + x \cos xy - x^2 y \sin xy - 2y \sin xy - xy^2 \cos xy \\ &= (2x - xy^2) \cos xy - (x^2 y + 2y) \sin xy \end{aligned} \quad (1)$$

$$f_y = x^2 \cos xy + \cos xy - xy \sin xy$$

$$\begin{aligned} f_{yx} &= 2x \cos xy - x^2 y \sin xy - y \sin xy - y \sin xy - xy^2 \cos xy \\ &= (2x - xy^2) \cos xy - (x^2 y + 2y) \sin xy \end{aligned} \quad (2)$$

It is clear from (1) and (2) that $f_{xy} = f_{yx}$

$$32. f(x, y) = \frac{xy}{\sqrt{1+x^2+y^2}}$$

$$\text{Sol. } f(x, y) = \frac{xy}{\sqrt{1+x^2+y^2}}$$

$$f_y = \frac{\partial f}{\partial y} = x \cdot \frac{\sqrt{1+x^2+y^2} \cdot 1 - y \cdot \frac{1}{2} \frac{2y}{\sqrt{1+x^2+y^2}}}{(\sqrt{1+x^2+y^2})^2}$$

$$= \frac{x}{1+x^2+y^2} \left[\frac{1+x^2+y^2-y^2}{\sqrt{1+x^2+y^2}} \right] = \frac{x(1+x^2)}{(1+x^2+y^2)^{3/2}}$$

$$f_{yx} = \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial x} \left[\frac{x(1+x^2)}{(1+x^2+y^2)^{3/2}} \right]$$

$$= \frac{(1+x^2+y^2)^{3/2} (1+3x^2) - x(1+x^2) \cdot \frac{3}{2} (1+x^2+y^2)^{1/2} \cdot 2x}{(1+x^2+y^2)^3}$$

$$= (1+x^2+y^2)^{1/2} \left[\frac{(1+x^2+y^2)(1+3x^2) - (x+x^3) \cdot 3x}{(1+x^2+y^2)^3} \right]$$

$$\begin{aligned}
 &= \frac{1}{(1+x^2+y^2)^{5/2}} [1+x^2+y^2+3x^2+3x^4+3x^2y^2-3x^2-3x^4] \\
 &= \frac{1+x^2+y^2+3x^2y^2}{(1+x^2+y^2)^{5/2}} \quad (1)
 \end{aligned}$$

Also

$$\begin{aligned}
 f_x &= \frac{\partial f}{\partial x} = y \cdot \frac{\sqrt{1+x^2+y^2} \cdot 1 - x \cdot \frac{1}{2} \frac{2x}{\sqrt{1+x^2+y^2}}}{(1+x^2+y^2)} \\
 &= \frac{y}{1+x^2+y^2} \cdot \frac{1+y^2}{\sqrt{1+x^2+y^2}} = \frac{y(1+y^2)}{(1+x^2+y^2)^{3/2}} \\
 f_{xy} &= \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial}{\partial y} \left[\frac{y(1+y^2)}{(1+x^2+y^2)^{3/2}} \right] \\
 &= \frac{(1+x^2+y^2)^{3/2} (1+3y^2) - y(1+y^2) \frac{3}{2} (1+x^2+y^2)^{1/2} \cdot 2y}{(1+x^2+y^2)^3} \\
 &= (1+x^2+y^2)^{1/2} \frac{[(1+x^2+y^2)(1+3y^2) - 3y^2(1+y^2)]}{(1+x^2+y^2)^3} \\
 &= \frac{[1+x^2+y^2+3x^2y^2]}{(1+x^2+y^2)^{5/2}} [1+x^2+y^2+3x^2y^2] \quad (2)
 \end{aligned}$$

From (1) and (2), we see that

$$f_{xy} = f_{yx}$$

Show that each of the following functions satisfies

Laplace's equation $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$. (Problems 33 – 36):

33. $f(x, y) = \sin x \sinh y$

Sol. $\frac{\partial f}{\partial x} = \cos x \sinh y$ and $\frac{\partial^2 f}{\partial x^2} = -\sin x \sinh y$ (1)

$\frac{\partial f}{\partial y} = \sin x \cosh y$ and $\frac{\partial^2 f}{\partial y^2} = \sin x \sinh y$ (2)

Adding (1) and (2), we have

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = -\sin x \sinh y + \sin x \sinh y = 0$$

34. $f(x, y) = e^{-x} \cos y$

Sol. $\frac{\partial f}{\partial x} = -e^{-x} \cos y$ and $\frac{\partial^2 f}{\partial x^2} = e^{-x} \cos y$ (1)

$\frac{\partial f}{\partial y} = e^{-x} (-\sin y) = -e^{-x} \sin y$ and $\frac{\partial^2 f}{\partial y^2} = -e^{-x} \cos y$ (2)

Adding (1) and (2), we have

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = e^{-x} \cos y - e^{-x} \cos y = 0$$

35. $f(x, y) = \ln \sqrt{x^2 + y^2}$

Sol. $f(x, y) = \ln \sqrt{x^2 + y^2} = \frac{1}{2} \ln (x^2 + y^2)$

$$\frac{\partial f}{\partial x} = \frac{1}{2} \cdot \frac{1}{x^2 + y^2} \cdot 2x = \frac{x}{x^2 + y^2}$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{(x^2 + y^2) \cdot 1 - x \cdot 2x}{(x^2 + y^2)^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2} \quad (1)$$

$$\frac{\partial f}{\partial y} = \frac{1}{2} \cdot \frac{1}{x^2 + y^2} \cdot 2y = \frac{y}{x^2 + y^2}$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{(x^2 + y^2) \cdot 1 - y \cdot 2y}{(x^2 + y^2)^2} = \frac{x^2 - y^2}{(x^2 + y^2)^2} \quad (2)$$

Adding (1) and (2), we get

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = \frac{y^2 - x^2}{(x^2 + y^2)^2} + \frac{x^2 - y^2}{(x^2 + y^2)^2} = 0$$

36. $f(x, y) = \arctan \left(\frac{2xy}{x^2 - y^2} \right)$

Sol. $f(x, y) = \arctan \left(\frac{2xy}{x^2 - y^2} \right)$

$$\frac{\partial f}{\partial x} = \frac{1}{1 + \frac{4x^2 y^2}{(x^2 - y^2)^2}} \times \frac{(x^2 - y^2) 2y - 2xy \cdot 2x}{(x^2 - y^2)^2}$$

$$= \frac{(x^2 - y^2)^2}{(x^2 + y^2)^2} \times \frac{-2y(x^2 + y^2)}{(x^2 - y^2)^2} = \frac{-2y}{x^2 + y^2}$$

$$\frac{\partial^2 f}{\partial x^2} = (-2y) (-1) (x^2 + y^2)^{-2} \cdot 2x = \frac{4xy}{(x^2 + y^2)^2} \quad (1)$$

$$\frac{\partial f}{\partial y} = \frac{1}{1 + \frac{4x^2 y^2}{(x^2 - y^2)^2}} \times \frac{(x^2 - y^2) 2x - 2xy \cdot (-2y)}{(x^2 - y^2)^2}$$

$$= \frac{(x^2 - y^2)^2}{(x^2 + y^2)^2} \times \frac{2x(x^2 + y^2)}{(x^2 - y^2)^2} = \frac{2x}{(x^2 + y^2)}$$

$$\frac{\partial^2 f}{\partial y^2} = 2x (-1) (x^2 + y^2)^{-2} \times 2y \frac{-4xy}{(x^2 + y^2)^2} \quad (2)$$

Adding (1) and (2), we see that

$$\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$$

37. $f(x, y) = x^2 \arctan\left(\frac{y}{x}\right) - y^2 \arctan\left(\frac{x}{y}\right)$, show that $\frac{\partial^2 f}{\partial x \partial y} = \frac{x^2 - y^2}{x^2 + y^2}$

Sol. $f(x, y) = x^2 \arctan \frac{y}{x} - y^2 \arctan \frac{x}{y}$

$$\frac{\partial f}{\partial y} = x^2 \cdot \frac{\partial}{\partial y} \left(\arctan \frac{y}{x} \right) + \arctan \frac{y}{x} \cdot \frac{\partial}{\partial y} (x^2) \\ - y^2 \cdot \frac{\partial}{\partial y} \left(\arctan \frac{x}{y} \right) + \arctan \frac{x}{y} \cdot \frac{\partial}{\partial y} (y^2)$$

$$= \frac{x^2 \left(\frac{1}{x} \right)}{1 + \left(\frac{y}{x} \right)^2} + \left(\arctan \frac{y}{x} \right) \cdot 0 - y^2 \frac{1 \left(-\frac{x}{y^2} \right)}{1 + \left(\frac{x}{y} \right)^2} - 2y \arctan \frac{x}{y}$$

$$= \frac{x}{x^2 + y^2} + \frac{x}{x^2 + y^2} - 2y \arctan \frac{x}{y}$$

$$= \frac{x^3}{x^2 + y^2} + \frac{xy^2}{x^2 + y^2} - 2y \arctan \frac{x}{y}$$

$$= \frac{x(x^2 + y^2)}{x^2 + y^2} - 2y \arctan \frac{x}{y} = x - 2y \arctan \frac{x}{y}$$

$$\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial}{\partial x} \left(x - 2y \arctan \frac{x}{y} \right) = 1 - 2y \frac{1}{1 + \left(\frac{x}{y} \right)^2} \times \frac{1}{y}$$

$$\text{or } \frac{\partial^2 f}{\partial x \partial y} = 1 - \frac{2}{1 + \frac{x^2}{y^2}} = 1 - \frac{2y^2}{x^2 + y^2} = \frac{x^2 + y^2 - 2y^2}{x^2 + y^2} = \frac{x^2 - y^2}{x^2 + y^2}$$

38. If $f(x, y) = \frac{x^2 + y^2}{x + y}$, that $(f_x - f_y)^2 = 4(1 - f_x - f_y)$

Sol. $f(x, y) = \frac{x^2 + y^2}{x + y}$

$$f_x = \frac{(x + y) \frac{\partial}{\partial x} (x^2 + y^2) - (x^2 + y^2) \left(\frac{\partial}{\partial x} \right) (x + y)}{(x + y)^2}$$

$$= \frac{(x + y) (2x) - (x^2 + y^2) \cdot 1}{(x + y)^2} = \frac{x^2 + 2xy - y^2}{(x + y)^2}$$

$$f_y = \frac{(x + y) 2y - (x^2 + y^2) \cdot 1}{(x + y)^2} = \frac{2xy + y^2 - x^2}{(x + y)^2}$$

$$f_x - f_y = \frac{x^2 + 2xy - y^2 - (2xy + y^2 - x^2)}{(x + y)^2}$$

$$= \frac{2x^2 - 2y^2}{(x+y)^2} = \frac{2(x^2 - y^2)}{(x+y)^2} = \frac{2(x-y)}{x+y}$$

$$(f_x - f_y)^2 = \frac{4(x-y)^2}{(x+y)^2}$$

$$\begin{aligned} \text{Again, } 1 - f_x - f_y &= 1 - \frac{x^2 + 2xy - y^2}{(x+y)^2} - \frac{2xy + y^2 - x^2}{(x+y)^2} \\ &= 1 - \left[\frac{2xy + 2xy}{(x+y)^2} \right] = 1 - \frac{4xy}{(x+y)^2} \\ &= \frac{(x+y)^2 - 4xy}{(x+y)^2} = \frac{(x-y)^2}{(x+y)^2} \end{aligned}$$

$$\text{Therefore, } 4(1 - f_x - f_y) = \frac{4(x-y)^2}{(x+y)^2}$$

$$\text{Hence } (f_x - f_y)^2 = 4(1 - f_x - f_y)$$

39. Show that the function $f(x, y) = \sin xy$ satisfies the differential equation $x^2 f_{xx} - y^2 f_{yy} = 0$.

Sol. $f(x, y) = \sin xy$

$$f_x = y \cos(xy) \quad \text{and} \quad f_{xx} = -y^2 \sin(xy)$$

$$f_y = x \cos(xy) \quad \text{and} \quad f_{yy} = -x^2 \sin(xy)$$

$$x^2 f_{xx} - y^2 f_{yy} = -x^2 y^2 \sin(xy) + y^2 x^2 \sin(xy) = 0 \text{ as required}$$

40. Let $f(x, y) = \begin{cases} x^2 \arctan\left(\frac{y}{x}\right) - y^2 \arctan\left(\frac{x}{y}\right) & \text{if } (x, y) \neq (0, 0) \\ 0 & \text{if } (x, y) = (0, 0) \end{cases}$

Show that $f_{xy}(0, 0) \neq f_{yx}(0, 0)$

Sol. $f_x(x, y) = x^2 \cdot \frac{1}{1 + \frac{y^2}{x^2}} \cdot \left(\frac{-y}{x^2}\right) + 2x \arctan \frac{y}{x} - y^2 \cdot \frac{1}{\left(1 + \frac{x^2}{y^2}\right)} \cdot \frac{1}{y}$

$$= \frac{-x^2 y}{x^2 + y^2} + 2x \arctan \frac{y}{x} - \frac{y^3}{x^2 + y^2}$$

$$= \frac{-y(x^2 + y^2)}{x^2 + y^2} + 2x \tan \frac{y}{x}$$

$$= 2x \arctan \frac{y}{x} - y$$

$$\text{Hence } f_x(0, h) = -h.$$

$$\text{Similarly, } f_y(x, y) = x - 2y \arctan \frac{x}{y}$$

$$\text{and so } f_y(h, 0) = h$$

$$f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0}{h} = 0$$

$$f_y(0, 0) = \lim_{k \rightarrow 0} \frac{f(0, k) - f(0, 0)}{k} = \lim_{k \rightarrow 0} \frac{0}{k} = 0$$

$$f_{xy}(0, 0) = \lim_{k \rightarrow 0} \frac{f_x(0, k) - f_x(0, 0)}{k} = \lim_{k \rightarrow 0} \frac{-k}{k} = -1$$

$$\text{And } f_{yx}(0, 0) = \lim_{h \rightarrow 0} \frac{f_y(h, 0) - f_y(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{h}{h} = 1$$

Thus $f_{xy}(0, 0) \neq f_{yx}(0, 0)$

41. (i) Let $f(x, y, z) = x^3 + 3yz + \sin xyz$. Prove that $f_{xyz} = f_{zxy}$

(ii) If $f(x, y, z, w) = \frac{xy}{z+w}$, show that $f_{xyzw} = \frac{2}{(x+w)^3}$

Sol.

(i) $f(x, y, z) = x^3 + 3yz + \sin xyz$

$$f_x = 3x^2 + yz \cos xyz$$

$$f_{xy} = z \cos xyz + yz (xz) (-\sin xyz) = z \cos xyz - xyz^2 \sin xyz$$

$$f_{xyz} = \cos xyz - xyz \sin xyz - 2xyz (\sin xyz) - x^2 y^2 z^2 (\cos xyz) \quad (1)$$

$$f_z = 3y + xy \cos xyz$$

$$f_{zx} = y \cos xyz - xy^2 z \sin xyz$$

$$f_{zxy} = \cos xyz - xyz \sin xyz - 2xyz \sin xyz - x^2 y^2 z^2 \cos xyz \quad (2)$$

The result follows from (1) and (2).

(ii) If $f(x, y, z, w) = \frac{xy}{z+w}$, show that $f_{xyzw} = \frac{2}{(x+w)^3}$

Sol. $f(x, y, z, w) = \frac{xy}{z+w}$

$$f_x = \frac{y}{z+w}$$

and

$$f_{xy} = \frac{1}{z+w}$$

$$f_{xyz} = \frac{-1}{(z+w)^2}$$

$$f_{xyzw} = \frac{0 - (-1) \cdot 2(z+w)}{(z+w)^4} = \frac{2}{(z+w)^3} \text{ as required.}$$

In Problems 42 – 45, find $\frac{dy}{dx}$ by using partial derivatives:

42. $y^2 + x^2y + ax^4 = 0$

Sol. Let $f(x, y) = y^2 + x^2y + ax^4 = 0$

$$\text{Now } f_x = 2xy + 4ax^3$$

and $f_y = 2y + x^2$

Therefore, $\frac{dy}{dx} = -\frac{f_x}{f_y} = -\frac{2xy + 4ax^3}{2y + x^2}$

43. $3x^2 - y^2 + x^3 = 0$

Sol. Let $f(x, y) = 3x^2 - y^2 + x^3 = 0$

Now $f_x = 6x + 3x^2$ and $f_y = -2y$

Therefore, $\frac{dy}{dx} = -\frac{f_x}{f_y} = -\frac{6x + 3x^2}{-2y} = \frac{6x + 3x^2}{2y}$

44. $x^2 + xy + y^2 + ax + by = 0$

Sol. Let $f(x, y) = x^2 + xy + y^2 + ax + by = 0$

Now $f_x = 2x + y + a$

and $f_y = x + 2y + b$

Thus $\frac{dy}{dx} = -\frac{f_x}{f_y} = -\frac{2x + y + a}{x + 2y + b}$

45. $x^3 + x^2 + xy^2 + \sin y = 0$

Sol. Let $f(x, y) = x^3 + x^2 + xy^2 + \sin y = 0$

Now, $f_x = 3x^2 + 2x + y^2$

$f_y = 2xy + \cos y$

Therefore, $\frac{dy}{dx} = -\frac{f_x}{f_y} = -\frac{3x^2 + 2x + y^2}{2xy + \cos y}$